



Assessing the forecast skill of a stochastic weather prediction model in predicting certain documented instances of extreme precipitation events across India

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Abstract

Extreme precipitation events are a serious threat, as they often lead to flooding, landslides, and have an overall detrimental impact on agriculture and livelihoods of people. Thus a reliable prediction of such events is very important for an effective risk management response. Since these events arise often from a very complex and inherently uncertain chain of atmospheric processes, probabilistic forecasting methods are preferred, as they allow us to quantify uncertainty and provide a more complete picture than a single output provided by deterministic forecasts. Here, we evaluate the performance of a stochastic machine-learning based weather prediction model – aifs-ens-1.0 – for ten documented extreme rainfall events across different regions of India. For each event, a 10-member ensemble was initialized five days before the event date. The analysis then evaluates the forecast against the corresponding ERA5 reanalysis precipitation fields for each event.

Background

An extreme precipitation event is defined statistically as an occurrence where the rainfall at a particular place and time exceeds some predefined threshold (usually the 95th or 99th percentile) of a long-term climatological record for that location.

Now, weather forecasting models can be broadly divided into two categories:

- **Deterministic:** Predicts a single, *most likely outcome*. It provides no information about forecast uncertainty.
- **Probabilistic:** Generates a range of possible outcomes and their likelihoods from an *ensemble* of model runs.

The atmosphere – being an inherently chaotic system – is very sensitive to even the minutest uncertainties in its initial state. So a single deterministic forecast, because of being constrained to just one set of initial conditions, may fail to properly account for instabilities that can later amplify, and result in the extreme event. Thus probabilistic forecasts in this case are superior, as they provide a way to measure the forecast uncertainty.

Ensemble construction

To generate an ensemble of future atmospheric states, there exist two broad approaches:

1. Perturbing the initial state (NWP)
2. Injecting stochasticity into a neural network (AI)

About AIFS-CRPS

The AIFS-CRPS (aifs-ens-1.0) is a data-driven model developed at the European Centre for Medium-Range Weather Forecasts (ECMWF) specifically for probabilistic weather prediction.

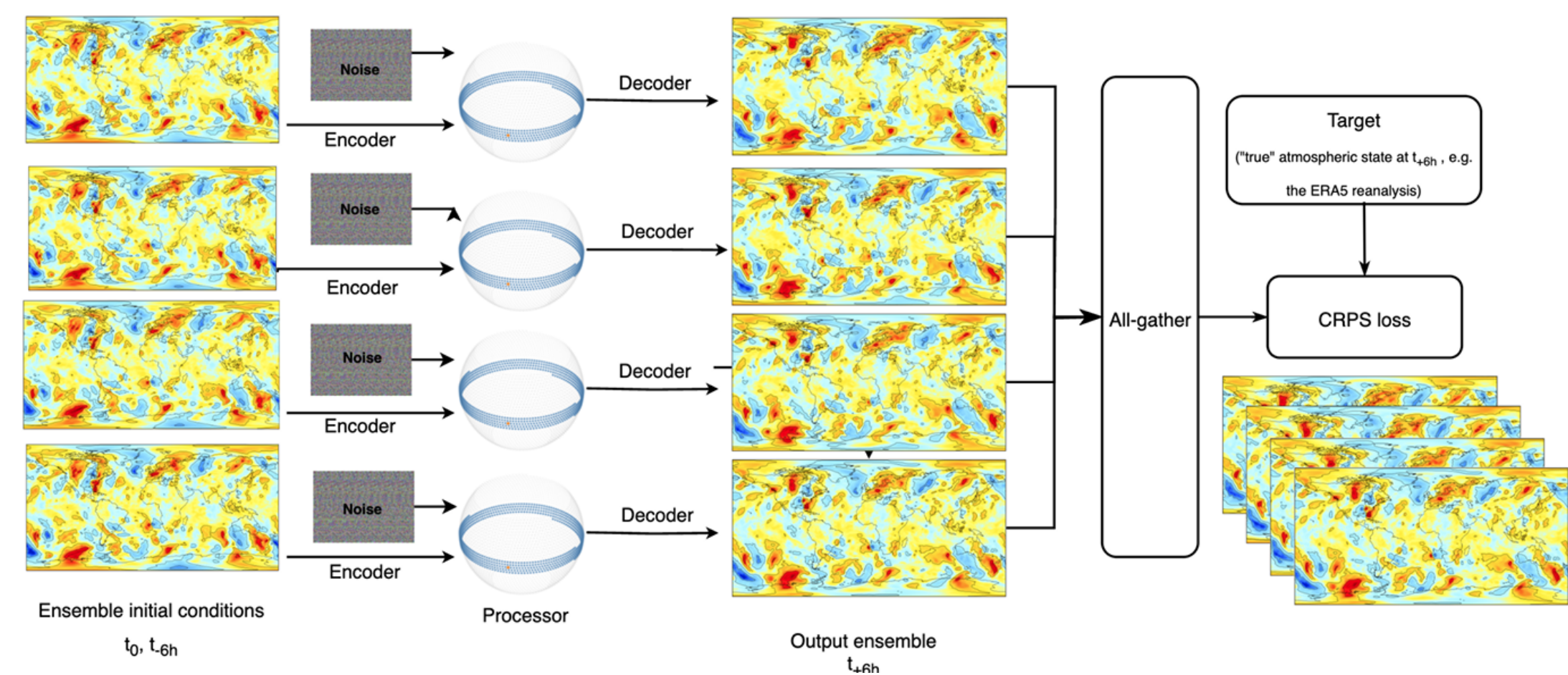


Figure 1: Probabilistic training of AIFS-CRPS

- **Training Objective & Loss Functions:** early AI models used MSE loss, this leads to over-smooth forecasts. AIFS-CRPS is trained on a refined probabilistic score.
- **The Continuous Ranked Probability Score (CRPS):**
$$CRPS(\{x_j\}, y) = \frac{1}{M} \sum_{j=1}^M |x_j - y| - \frac{1}{2M^2} \sum_{j=1}^M \sum_{k=1}^M |x_j - x_k|$$
where $\{x_j\}$ is the M-member ensemble forecast and y is the verifying observation.
- **The Fair CRPS (fCRPS):** The standard CRPS is statistically biased for finite ensemble sizes. The fCRPS corrects for this.
$$fCRPS(\{x_j\}, y) = \frac{1}{M} \sum_{j=1}^M |x_j - y| - \frac{1}{2M(M-1)} \sum_{j=1}^M \sum_{k \neq j}^M |x_j - x_k|$$

However, fCRPS suffers from a *degeneracy* issue – if all but one member perfectly match the truth, the loss for the final member will become zero, irrespective of its value.

3. The Almost Fair CRPS (afCRPS): To get the best of both worlds – most of the fairness of fCRPS and the stability of CRPS – the model is trained on a weighted average of the two.

$$afCRPS := \alpha \cdot fCRPS + (1 - \alpha) \cdot CRPS$$

- **Architecture & Data:** It uses a transformer-based graph neural network (GNN) and was trained on the Copernicus ERA5 global reanalysis dataset from 1979-2017.

- **Input Variables:** The model is fed a comprehensive snapshot of the atmosphere:

Field / Variable	Level type	Input/Output	Normalisation
Geopotential, horizontal and vertical wind components, specific humidity, temperature	Pressure level: 50, 100, 150, 200, 250, 300, 400, 500, 600, 700, 850, 925, 1000	Both	Standardised, apart from the geopotential, which is max-scaled
Surface pressure, mean sea-level pressure, skin temperature, 2 m temperature, 2 m dewpoint temperature, 10 m horizontal wind components, total column water	Surface	Both	Standardised
Total precipitation	Surface	Output	Standard deviation changed but mean kept the same
Land-sea mask, orography, standard deviation and slope of sub-grid orography, insolation, cosine and sine of latitude and longitude, cosine and sine of the local time of day and day of year	Surface	Input	Sub-grid orography max-scaled; all others not normalised
Time step identifier	-	Input	-

Figure 2: Input and output variables of AIFS-CRPS

Data

Ten historical extreme rainfall events were selected from the XWeatherLib repository.

Serial No.	Region	Date of Event
1	Maharashtra	2002-08-25
2	Guwahati	2004-04-03
3	Odisha	2007-08-06
4	Uttarakhand	2013-07-14
5	Kashmir	2014-09-04
6	Chennai	2015-12-01
7	Yamunanagar	2016-07-15
8	Kerala	2018-08-14
9	Kerala	2019-08-08
10	Telangana	2023-03-18

Figure 3: something

Methodology and Analysis

1. Forecasting and evaluation protocol

For each event, a 10-member AIFS-CRPS ensemble was initialized 5 days prior using ERA5 data and verified against ERA5 ground truth. Performance at an extreme (90th percentile) threshold was measured by the Critical Success Index (CSI):

$$CSI = \frac{H}{H + M + F}$$

where H, M, and F are the counts of Hits, Misses, and False alarms, respectively.

2. Performance summary

The skill and consensus of the 10-member ensemble varied significantly across events as summarized below.

Event Location	Date	Skillful Members (out of 10)	Members with Some Skill (out of 10)	Failed Members (out of 10)	Ensemble Consensus & Confidence
Yamunanagar	2016-07-15	10	0	0	Extremely High Confidence
Chennai	2015-12-01	10	0	0	Extremely High Confidence
Maharashtra	2002-08-25	7	3	0	High Confidence
Kerala	2019-08-08	10	0	0	High Confidence
Kashmir	2014-09-04	1	1	8	Very Low Confidence
Guwahati	2004-04-03	1	0	9	Very Low Confidence
Telangana	2023-03-18	1	2	7	Low Confidence
Uttarakhand	2013-07-14	0	7	3	Low Confidence
Kerala	2018-08-14	0	0	10	Complete Forecast Failure
Odisha	2007-08-06	0	0	10	Complete Forecast Failure

Figure 4: Caption

Plots

Four variables were found to be strongly correlated with extreme rainfall: **Total Column Water (TCW)**, **Specific Humidity (q_{850})**, **Vertical Velocity (w_{500})**, and **Temperature (t_{700})**.

(Telangana Case)

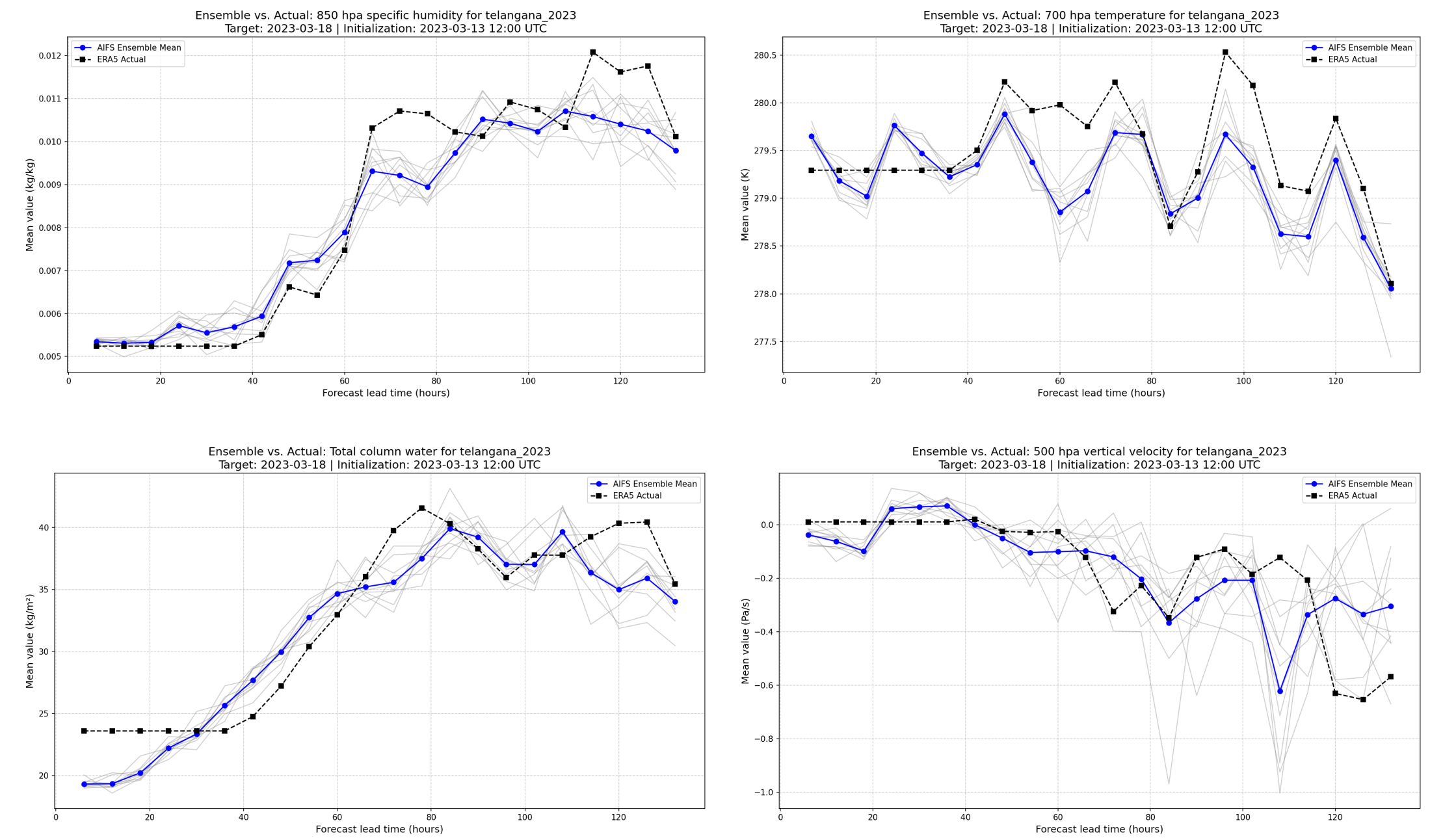


Figure 5: Time series of key atmospheric precursors for the Telangana event.

2. Precipitation Forecasts: Three Examples

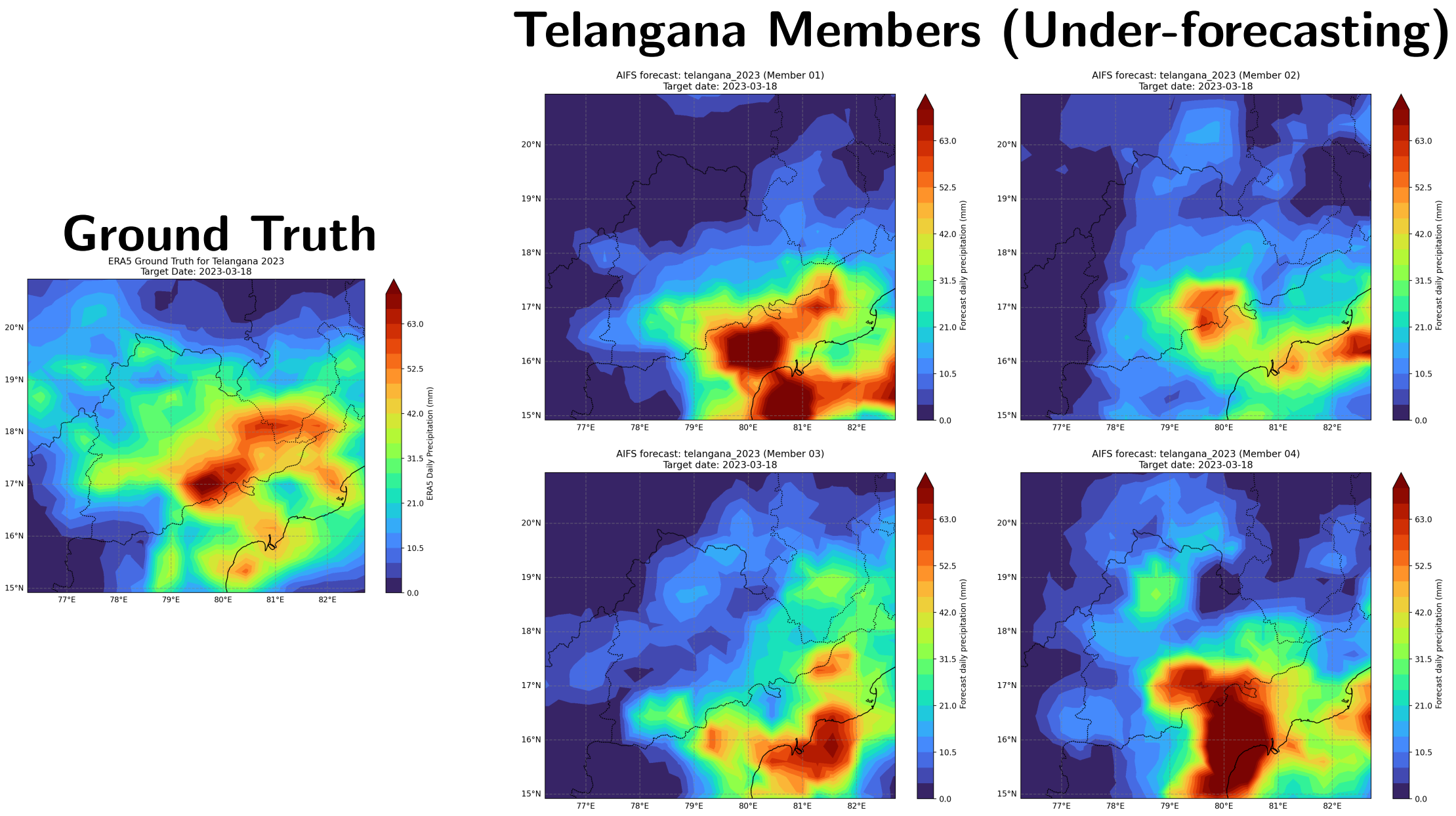


Figure 6: Under-forecasting: Correct location, underestimated intensity.

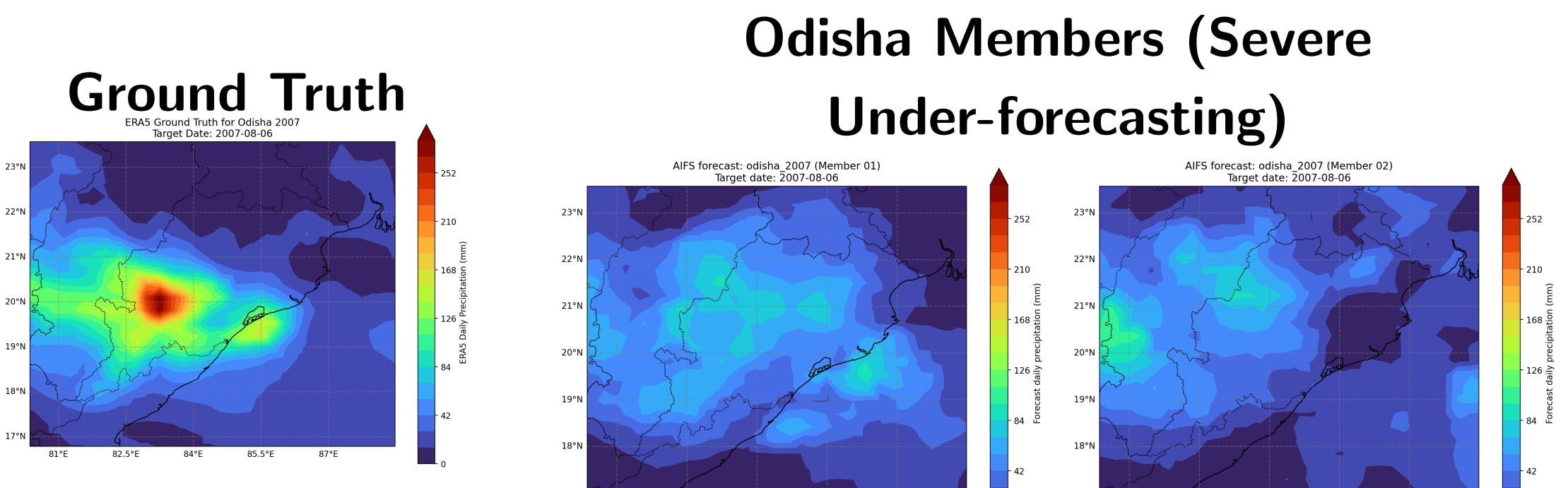


Figure 7: Severe Under-forecasting: Model missed the event's magnitude.

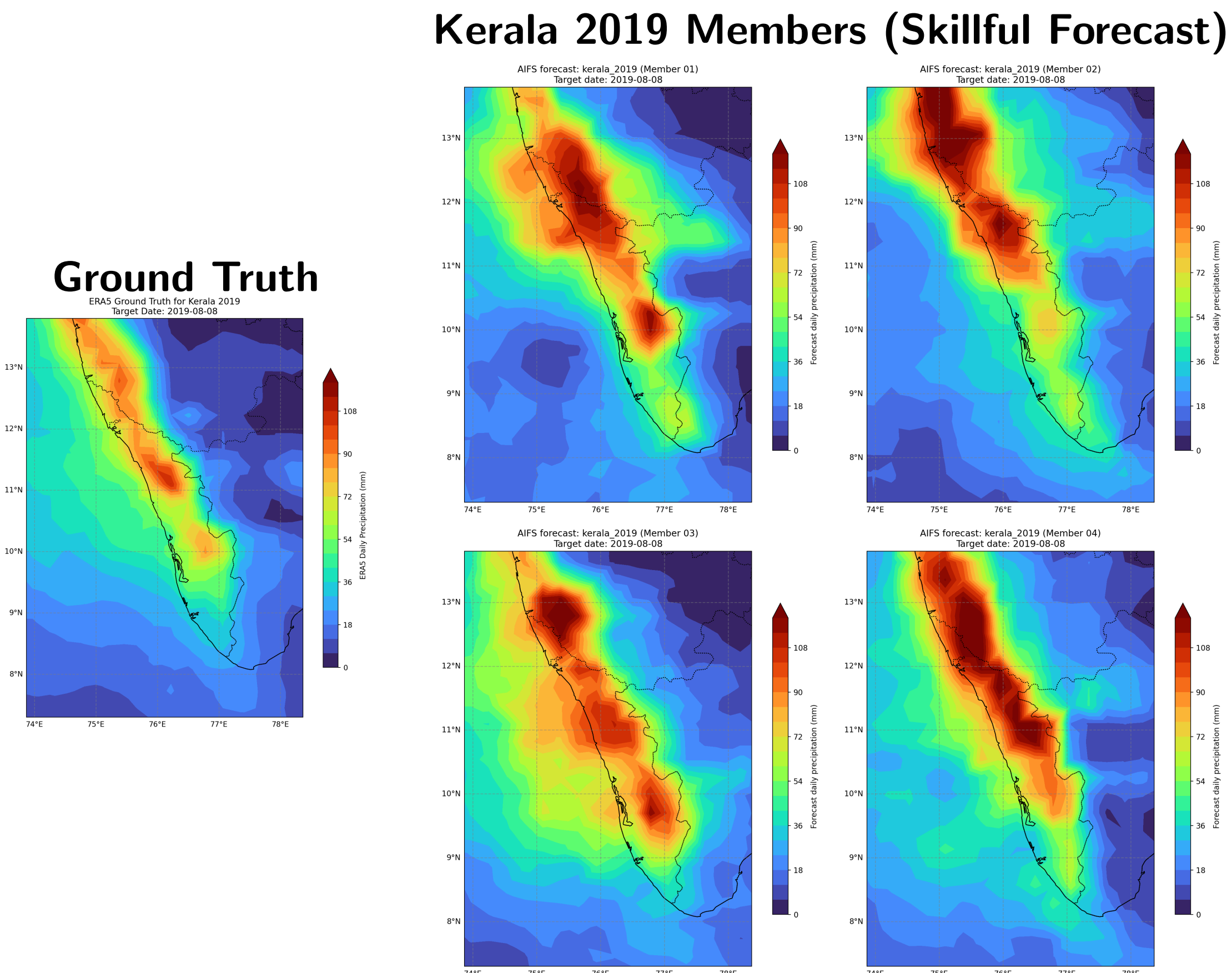


Figure 8: Skillful Forecast: Correct location and intensity captured.

References

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